Getting Started

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Instructions

This document runs a simple analysis of the Table 1 from (Wansink and Payne 2009). Edit the header information to show your name and the date you complete the assignment.

Typeset this document in RStudio. You should not need to install additional R libraries.

When you are able to typeset the original document on your computer, modify the code to analyze Calories per Serving. That is, replace all occurrences of the variable names of the form `CaloriesPerServing*` to the variable `CaloriesPerServing*`. Change the values in the table named `ReferenceValues` to match the values in the text of Wansink and Payne (Wansink and Payne 2009).

Typeset the edited document and analyze the results. Add your own comments to the section headed **Comments**. Format your comments as *italicized* text, **bold** text or

> as quoted text.

Change the name of this file to match your user name on D2L, keeping the 'Rmd' extension, and include week number and coding language in the title (for example, `Peter.Claussen.R.1.Rmd`). Upload this file to D2L. Typeset this file to Word or PDF and upload the result to D2L as well.

Data

Table 1: Mean and (SD) for selected recipes from "Joy of Cooking"

Measure	1936	1946	1951	1963	1975	1997	2006
calories per recipe (SD)	2123.8 (1050.0)	2122.3 (1002.3)	2089.9 (1009.6)	2250.0 (1078.6)	2234.2 (1089.2)	2249.6 (1094.8)	3051.9 (1496.2)
calories per serving (SD)	268.1 (124.8)	271.1 (124.2)	280.9 (116.2)	294.7 (117.7)	285.6 (118.3)	288.6 (122.0)	384.4 (168.3)
servings per recipe (SD)	12.9 (13.3)	12.9 (13.3)	13.0 (14.5)	12.7 (14.6)	12.4 (14.3)	12.4 (14.3)	12.7 (13.0)

Analysis

Enter data

```
CookingTooMuch.dat <- data.frame(
  Year=c(1936, 1946, 1951, 1963, 1975, 1997, 2006),
  CaloriesPerServingMean = c(2123.8, 2122.3, 2089.9, 2250.0, 2234.2, 2249.6, 3051.9),
  CaloriesPerServingSD = c(1050.0, 1002.3, 1009.6, 1078.6, 1089.2, 1094.8, 1496.2),
  CaloriesPerServingMean = c(268.1, 271.1, 280.9, 294.7, 285.6, 288.6, 384.4),
  CaloriesPerServingSD = c(124.8, 124.2, 116.2, 117.7, 118.3, 122.0, 168.3),
  ServingsPerRecipeMean = c(12.9, 12.9, 13.0, 12.7, 12.4, 12.4, 12.7),
  ServingsPerRecipeSD = c(13.3, 13.3, 14.5, 14.6, 14.3, 14.3, 13.0)
)
```

Create values for confidence interval plot

Wansink reports that 18 recipes were analyzed.

```
n <- 18
```

Use standard formula for standard error $\widehat{se} = \widehat{\sigma}/\sqrt{n}$ and confidence interval, assuming a significance level α of 5%.

$$CI(95\%) = \widehat{\mu} \pm z_{\alpha/2} \times \widehat{se}$$

.

```
StandardError <- function(sigma, n) {
  sigma/sqrt(n)
}
ConfidenceBound <- function(sigma, n, alpha = 0.05) {
  qnorm(1-alpha/2)*StandardError(sigma,n)
}
```

Create a variable for plotting and calculate upper and lower bounds using confidence intervals.

```
PlotCookingTooMuch.dat <- CookingTooMuch.dat
PlotCookingTooMuch.dat$CaloriesPerServing <- PlotCookingTooMuch.dat$CaloriesPerServingMean
PlotCookingTooMuch.dat$Lower <- PlotCookingTooMuch.dat$CaloriesPerServing - ConfidenceBound(CookingTooMu
PlotCookingTooMuch.dat$Upper <- PlotCookingTooMuch.dat$CaloriesPerServing + ConfidenceBound(CookingTooMu
PlotCookingTooMuch.dat <- PlotCookingTooMuch.dat[,c("Year","CaloriesPerServing","Lower","Upper")]
```

Examine the values to make sure we've entered correctly.

```
print(PlotCookingTooMuch.dat)
```

```
##   Year CaloriesPerServing    Lower    Upper
## 1 1936             2123.8 1638.734 2608.866
## 2 1946             2122.3 1659.270 2585.330
## 3 1951             2089.9 1623.497 2556.303
## 4 1963             2250.0 1751.721 2748.279
## 5 1975             2234.2 1731.025 2737.375
## 6 1997             2249.6 1743.837 2755.363
## 7 2006             3051.9 2360.704 3743.096
```

Check our calculations. Wansink reports 95% confidence intervals for the 1936 and 2006 means as [210.4, 325.8] and [325.8, 514.7], respectively. We should be equal, to within rounding error.

```
CompValues <- PlotCookingTooMuch.dat[c(1,7),c("Lower","Upper")]
ReferenceValues <- matrix(c(210.4,325.8,359.1,514.7),nrow=2,byrow=TRUE)
CompValues
```

```
##      Lower    Upper
## 1 1638.734 2608.866
## 7 2360.704 3743.096
```

```
ReferenceValues
```

```
##       [,1]  [,2]
## [1,] 210.4 325.8
## [2,] 359.1 514.7
```

```
any(abs(CompValues-ReferenceValues)>0.1)
```

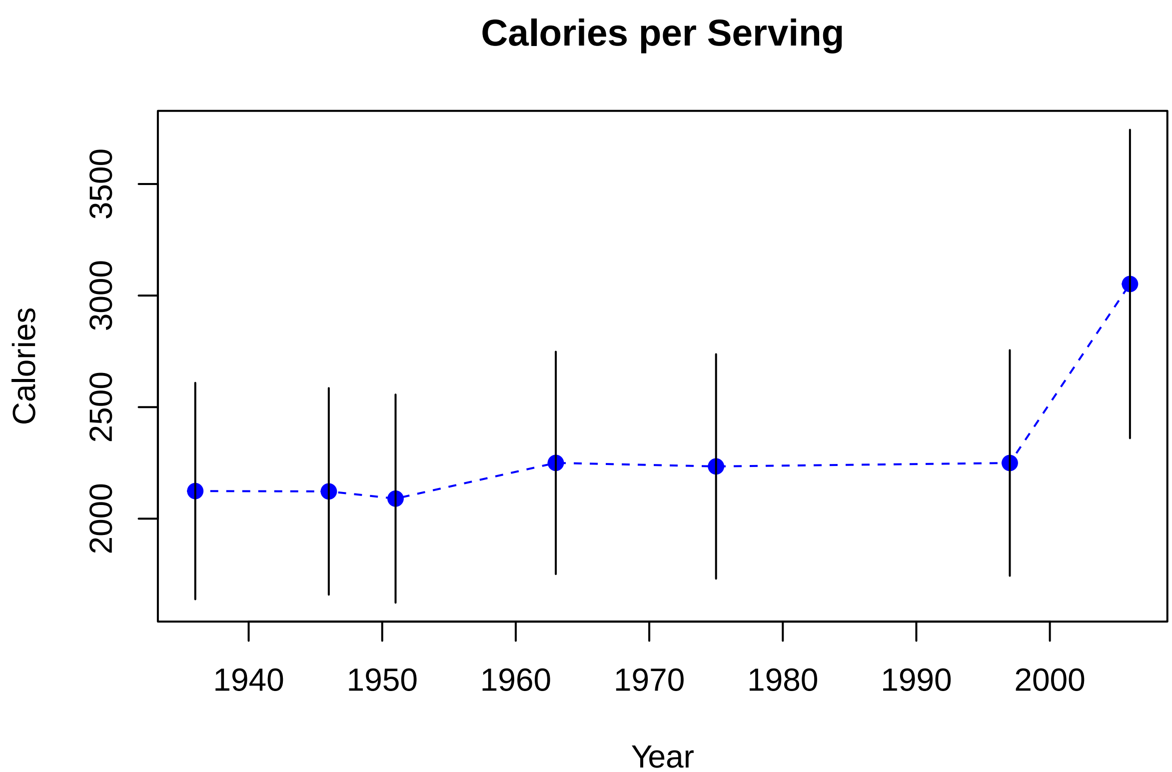
```
## [1] TRUE
```

We no longer need the original data.

```
CookingTooMuch.dat <- NULL
```

Plot the table

```r
plot(CaloriesPerServing ~ Year, data=PlotCookingTooMuch.dat,
     col="blue", pch=19,
     main="Calories per Serving",
     ylab="Calories",
     ylim=c(min(PlotCookingTooMuch.dat$Lower), max(PlotCookingTooMuch.dat$Upper)))
lines(CaloriesPerServing ~ Year, data=PlotCookingTooMuch.dat,
      lty="dashed", col="blue", lend=2)
segments(x0=PlotCookingTooMuch.dat$Year,
         y0=PlotCookingTooMuch.dat$Lower,
         x1=PlotCookingTooMuch.dat$Year,
         y1=PlotCookingTooMuch.dat$Upper)
```



Comments

From this plot, it appears that average calories per ***serving*** changed very little from 1936 to 1997. This suggests that the significant increase from 1936 to 2006 reported by Wansink is due to an increase in the 2006 recipes only. Thus, we only need to compare 1936 and 2006 recipe sets.

References

Wansink, Brian, and Collin Payne. 2009. "The Joy of Cooking Too Much: 70 Years of Calorie Increases in Classic Recipes" 150 (150): 291–92.